How Much Shorter Can a Prediction-Powered Confidence Interval Be? A Headroom Analysis with Fixed Labels, Unlabeled Data and Predictor

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Abstract

Prediction-powered inference (PPI) and its power-tuned variant PPI++ combine a small labeled sample, a large unlabeled sample and a fixed pretrained predictor into valid confidence intervals. We ask a narrow question: *with all three inputs held fixed and nominal coverage required, how much shorter can the interval for a population mean still get, and when?* We give an exact two-sample variance formula for the augmented estimator $\hat{\theta}_g$ (any fixed g of what is observed on every unit), show that the optimally power-tuned PPI++ interval is exactly the best *affine-in-the-prediction* member of this family, and derive the remaining headroom in closed form: it equals the variance of the part of $\mathbb{E}[Y \mid \text{observables}]$ that is not affine in the prediction, scaled by $N/(n(n+N))$. For regular estimators this is the whole headroom (semiparametric efficiency bound), so the analysis also says when nothing more can be gained. We check this on nine public datasets (ecology, astronomy, census, proteomics, elections, genomics; 1000 replications per cell) and eight synthetic designs with known truth, comparing PPI, PPI++, a classical interval and cross-fitted augmented estimators with affine, isotonic, covariate-linear and boosted-tree calibration. On all nine real datasets the estimated headroom is at most 2.4% of the variance and the best cross-fitted interval is within $[0.967, 1.011]$ of the PPI++ width at $n = 200$ labels (within $[0.996, 1.002]$ at $n = 1000$); the only visible gain ($\approx$ 3–7%) comes from a predictor whose slope exceeds one. On synthetic designs where the predictor is non-linearly calibrated or ignores informative covariates, widths fall to 0.49–0.61 of PPI++ and match the formula. The larger practical issue is validity: Wald intervals of all methods under-cover at small n (e.g. PPI++ 76% at $n = 25$ on census income), a studentized bootstrap repairs only part of it, and under covariate shift between labeled and unlabeled data the power-tuned intervals fail completely (0% coverage) while $\lambda = 1$ PPI does not.

1 Introduction

In many sciences gold-standard labels are scarce, while features and a pretrained predictor are cheap. Prediction-powered inference [Angelopoulos et al., 2023a] uses the predictor on a large unlabeled sample and corrects its bias with a small labeled sample, yielding intervals that are valid regardless of predictor quality and shorter than labeled-only intervals when the predictor is good. PPI++ [Angelopoulos et al., 2023b] adds a data-driven weight λ on the predictions so the interval never gets much worse than the classical one. Subsequent work changes the inputs: cross-fitting

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when the predictor is trained on the labels [Zrnic and Candès, 2023], choosing which points to label [Zrnic and Candès, 2024], stratification [Fisch et al., 2024].

We study the complementary question in which *none of the inputs may change*: the labeled set, the unlabeled set and the predictor are given, and only the way they are combined is ours to choose. Is the PPI++ interval already as short as a valid interval can be? Our contributions are:

1. **A closed-form headroom (Section 3).** An exact two-sample variance for the augmented estimator, the identification of optimally tuned PPI++ with its best affine member, and the exact gap to the efficient estimator: $V_{\mathrm{PPI++}} - V_{\mathrm{eff}} = \frac{N}{n(n+N)}\mathrm{Var}(\mu(Z) - b^\star F)$. The gap is zero when the calibration curve is affine and no covariate carries information the predictor lacks, and it vanishes as $N/n \to 0$. Most ingredients are classical (missing-data and semi-supervised mean estimation [Robins et al., 1994, Zhang et al., 2019]); the point is the exact comparison with PPI++ and the use of it as a diagnostic.
2. **A measurement of the headroom on real data (Section 5.1).** For all nine datasets in the PPI reference package the headroom for the mean is $\leq 2.4\%$ of the variance, and cross-fitted estimators do not shorten intervals appreciably; we report where the small gains are.
3. **Conditions under which large gains exist and are realised (Section 5.2).** Synthetic designs with a non-linear calibration curve or an informative covariate yield intervals 40–50% shorter, matching the formula once n is moderate; the price at small n is quantified.
4. **Validity caveats (Sections 5.3,5.4).** Wald intervals of every method are anti-conservative at small n on real data; shorter is not better unless coverage is checked. Under labeled/unlabeled shift, the gains that come from fitting calibration on labeled data turn into complete failure.

The honest summary is conditional: for a population mean with a reasonable predictor, PPI++ leaves little to gain; there is room only when the predictor is miscalibrated beyond an affine map or leaves covariate information unused.

2 Related Work

PPI family. PPI [Angelopoulos et al., 2023a] and PPI++ [Angelopoulos et al., 2023b] give asymptotically valid intervals for means, quantiles and M-estimands; PPI++ tunes a scalar (or matrix) weight on the predictor. Cross-PPI [Zrnic and Candès, 2023] handles predictors trained on the labeled data by cross-fitting; active inference [Zrnic and Candès, 2024] changes which units are labeled; stratified PPI [Fisch et al., 2024] applies PPI within strata of the predictor score, which corresponds to a piecewise-constant g in our notation. These change inputs or use particular calibration classes; we instead ask what the best use of fixed inputs could achieve and measure it.

Semiparametric efficiency, semi-supervised and post-prediction inference. Estimating a mean when outcomes are missing completely at random is a classical semiparametric problem whose efficient estimators are augmented inverse-probability-weighted [Robins et al., 1994, Bickel et al., 1993, Tsiatis, 2006], with semi-supervised mean estimation developed in Zhang et al. [2019]. Machine-learned nuisance functions are handled by cross-fitting [Chernozhukov et al., 2018]. POP-Inf [Miao et al., 2023] proposes an assumption-lean, data-adaptive post-prediction estimator that is at least as efficient as the classical and PPI estimators. Our estimators are standard members of this family; our contribution is the exact PPI++ comparison, the headroom decomposition and its empirical evaluation (including finite-sample validity and shift), not a new estimator.

3 Setting and a Headroom Theory

Setting. A predictor f is fixed. Every unit has observables $Z = (F, W)$ with $F = f(\text{features})$ the prediction and W optional covariates (empty in some datasets). We observe n labeled units $(Z_i, Y_i) \overset{iid}{\sim} P$ and N unlabeled units $Z'_j \overset{iid}{\sim} P_Z$, independent. The target is $\theta = \mathbb{E}[Y]$; $\mu(Z) = \mathbb{E}[Y \mid Z]$. We choose the mean because it is the canonical PPI target, has a closed-form efficiency theory, and every dataset has a natural mean (Section 4). The assumption that labeled and unlabeled units are exchangeable (A1: the i.i.d. sampling just described) is what the validity of all methods rests on; Section 5.4 shows what happens without it.

For a fixed function g with $\mathbb{E}g(Z)^2 < \infty$ define the augmented estimator

$$\hat{\theta}_g = \frac{1}{n+N}\Big(\sum_{i\le n} g(Z_i) + \sum_{j\le N} g(Z'_j)\Big) + \frac{1}{n}\sum_{i\le n}\big(Y_i - g(Z_i)\big). \tag{1}$$

Proposition 1 (exact variance). *$\hat{\theta}_g$ is unbiased and* $\mathrm{Var}(\hat{\theta}_g) = \dfrac{\mathrm{Var}(Y)}{n+N} + \dfrac{N}{n(n+N)}\mathrm{Var}\big(Y - g(Z)\big)$. *The minimum over g is attained at $g = \mu$ (up to constants):* $V_{\mathrm{eff}} = \frac{\mathrm{Var}(Y)}{n+N} + \frac{N}{n(n+N)}\mathbb{E}\mathrm{Var}(Y \mid Z)$.

Proposition 2 (PPI++ is the best affine member). *For $\lambda \in \mathbb{R}$ let* $\hat{\theta}^{\mathrm{PPI}}_{\lambda} = \frac{1}{n}\sum_i (Y_i - \lambda F_i) + \frac{\lambda}{N}\sum_j F'_j$, *with variance* $\mathrm{Var}(Y - \lambda F)/n + \lambda^2 \mathrm{Var}(F)/N$. *Its minimum over λ (attained at* $\lambda^\star = \mathrm{Cov}(Y,F)/(\mathrm{Var}(F)(1 + n/N))$*) equals* $\min_{a,b}\mathrm{Var}(\hat{\theta}_{a+bF}) =: V_{\mathrm{PPI++}}$, *and*

$$V_{\mathrm{PPI++}} - V_{\mathrm{eff}} = \frac{N}{n(n+N)}\,\mathrm{Var}\big(\mu(Z) - b^\star F\big), \qquad b^\star = \mathrm{Cov}(Y,F)/\mathrm{Var}(F). \tag{2}$$

Corollary 1 (headroom and its absence). *Let* $\kappa = \mathrm{Var}(\mu(Z) - b^\star F)/\mathrm{Var}(Y - b^\star F) \in [0,1]$ *(the share of the residual variance of the best affine calibration that Z can still explain). Then* $V_{\mathrm{eff}}/V_{\mathrm{PPI++}} \to 1 - \kappa$ *as $N/n \to \infty$, so the interval-width ratio is $\ge \sqrt{1-\kappa}$ asymptotically. (i) If $\mu(Z)$ is affine in F (calibrated predictor, no informative covariates), $\kappa = 0$ and PPI++ is already efficient. (ii) For finite N/n the gap carries the factor $N/(n+N)$; at $N = 0$ all methods reduce to the classical interval. (iii) If $\lambda^\star \notin [0,1]$, the clipped version used in* `ppi_py` *is strictly worse than $V_{\mathrm{PPI++}}$ and κ understates the headroom over it.*

Proofs are in Appendix A. Under the model above (nonparametric P, labels missing completely at random) V_{eff} is the semiparametric efficiency bound [Robins et al., 1994, Tsiatis, 2006], hence *no regular asymptotically linear estimator, and so no Wald interval built from one, can have smaller asymptotic variance*; the experiments in Section 5.2 confirm the formula and not just the bound. We do not claim a non-asymptotic lower bound on the width of arbitrary valid intervals.

Feasible estimator. Since μ is unknown we fit $\hat{g}$ on the labeled data with K-fold cross-fitting: labeled point i in fold k uses $\hat{g}^{(-k)}$ (trained only on labeled points outside fold k) in both terms of (1); unlabeled points use the average of the K fold models. The variance is estimated from the two samples separately, $\hat{V} = n\,\widehat{\mathrm{Var}}_i\big[\frac{Y_i - \hat{g}_i}{n} + \frac{\hat{g}_i}{n+N}\big] + \frac{N}{(n+N)^2}\widehat{\mathrm{Var}}_j[\hat{g}(Z'_j)]$, and the interval is $\hat{\theta} \pm z_{1-\alpha/2}\sqrt{\hat{V}}$.

Proposition 3 (validity, sketch). *If $\|\hat{g}^{(-k)} - g^*\|_{L_2(P_Z)} \to_p 0$ for some fixed g^* (not necessarily μ) and $\mathrm{Var}(Y - g^*) > 0$, then as $n, N \to \infty$ with N/n bounded away from 0, $(\hat{\theta} - \theta)/\sqrt{\hat{V}} \Rightarrow \mathcal{N}(0,1)$ and the asymptotic variance is that of Proposition 1 with $g = g^*$.*

Hence validity does not require $\hat{g}$ to be consistent for μ; consistency only buys efficiency. All of this is asymptotic, and says nothing about the finite-sample cost of estimating $\hat{g}$, which we measure.

4 Methods Compared and Protocol

Methods. (1) *classical*: labeled-only CLT interval; (2) *PPI* ($\lambda = 1$) and (3) *PPI++* (power-tuned, λ clipped to $[0,1]$), both from `ppi-python` 0.2.3 (`ppi_mean_ci`, which uses standard deviations with $\mathrm{ddof} = 0$; we left it unchanged as the reference implementation); (4) *AIPW-lin*: cross-fitted $\hat{g} = a + bF$; (5) *AIPW-iso*: isotonic regression of Y on F; when covariates exist also *AIPW-linX* (linear in (F, W)) and *AIPW-gbm* (histogram gradient boosting, depth 3, 60 trees [Pedregosa et al., 2011]); (6) *AIPW-auto*: among the above, the learner with the smallest out-of-fold squared error (all candidates share folds; selection uses the same out-of-fold predictions that enter the estimator). $K = 5$. All intervals are 95% two-sided Wald intervals unless stated; Section 5.3 adds studentized-bootstrap versions of classical, PPI++ and AIPW-lin.

Populations and sampling. Each file of the PPI reference package (data version `v0.2.3`) is treated as the population; labeled and unlabeled samples are drawn i.i.d. *with replacement*, so A1 holds exactly and the truth $\theta = \bar{Y}$ over the file is exact. Datasets (rows of Table 1): galaxies, forest, alphafold (covariates: three modification indicators), gene_expression, census_income (covariates: age, sex), census_healthcare, census_education, plankton (2013 half only here), ballots (the 1027 labeled ballots).

Table 1: Coverage (%) and width relative to PPI++ at $n = 200$, $N = 5000$, 1000 replications. b: OLS slope of Y on F; $\hat{\kappa}$: estimated headroom share (Cor. 1); bound $= \sqrt{1-\hat{\kappa}}$ is the asymptotic ($N/n \to \infty$) width ratio of the efficient estimator *relative to unclipped-λ PPI++*. Ratio < 1 means shorter than PPI++. Rows S· are synthetic. Cov. s.e. ≈ 0.7.

				coverage (%)					width / PPI++			
dataset	b	$\hat{\kappa}$	bound	clas.	PPI	PPI++	lin	auto	clas.	PPI	lin	auto
galaxies	0.98	0.000	1.000	94.1	93.0	92.5	93.4	92.8	1.456	1.001	1.008	1.006
forest	0.90	0.000	1.000	95.4	95.9	95.2	95.8	95.5	1.176	1.004	1.012	1.005
alphafold	1.03	0.008	0.996	94.8	95.4	95.6	95.3	94.2	1.657	0.995	1.005	0.998
gene_expression	1.15	0.024	0.988	94.5	94.9	94.6	95.0	94.7	2.935	0.998	0.967	0.967
census_income	1.38	0.011	0.994	92.4	90.7	90.6	90.7	90.6	1.237	1.000	1.002	0.995
census_healthcare	0.98	0.001	0.999	96.0	95.4	95.4	95.3	95.2	1.390	1.001	1.010	1.007
census_education	0.97	0.001	1.000	93.6	94.6	94.2	94.4	94.3	1.266	1.003	1.014	1.011
plankton	0.84	0.000	1.000	95.5	95.0	94.2	95.5	95.5	1.739	1.039	1.004	1.004
ballots	0.98	0.000	1.000	94.9	94.1	93.9	93.9	93.9	3.745	1.019	1.000	1.000
S0_calibrated	1.00	0.000	1.000	94.6	93.5	93.3	93.7	93.7	1.387	1.000	1.010	1.010
S1_nonlinear_link	1.66	0.884	0.341	94.3	92.7	92.7	92.4	90.4	1.396	1.000	0.953	0.550
S2_omitted_linear	1.00	0.798	0.449	94.3	94.7	94.5	94.6	94.7	1.320	1.001	1.011	0.509
S3_omitted_nonlinear	1.00	0.806	0.441	94.0	95.5	95.3	95.4	94.9	1.297	1.001	1.011	0.605
S4_noise_predictor	-0.00	0.800	0.447	95.4	95.4	95.2	95.4	96.4	1.001	1.356	1.012	0.486
S5_all_noise	-0.01	0.000	1.000	94.9	94.4	95.0	95.1	94.9	1.001	1.235	1.012	1.011
S6_heavy_tail	1.00	0.000	1.000	95.9	95.4	95.2	95.4	95.3	1.159	1.001	1.012	1.011
S7_binary_miscal	1.49	0.013	0.993	94.0	95.2	95.2	95.3	94.7	1.228	1.000	0.983	0.980

Synthetic designs S0–S7 (Appendix B) have analytic truth. Grid: $n \in \{25, 50, 100, 200, 500, 1000\}$ with $N = 5000$, and $N \in \{50, 200, 1000, 5000, 20000\}$ with $n = 200$; **1000 replications per cell**, seeds derived from `SeedSequence`$([20261101, \text{dataset}, n, N, \text{rep}])$, all methods evaluated on the same samples. Coverage has Monte-Carlo s.e. of about 0.7 points near 95%. The width ratio is mean width over mean PPI++ width on the same replications, with a paired bootstrap interval over replications (always $\leq \pm 0.002$ for ratios near one at $n \geq 200$ and reported in the code outputs); it reflects Monte-Carlo, not dataset, uncertainty. Headroom κ is estimated on the full population with 5-fold cross-validated learners (affine, isotonic, linear-in-(F, W), boosting); since the learner class is finite, true headroom could be somewhat larger than $\hat{\kappa}$.

5 Results

5.1 Real data: almost no headroom for the mean

Table 1 (top) shows that at $n = 200$ the best cross-fitted interval is between 0.967 and 1.011 of the PPI++ width on all nine real datasets (median 1.004); at $n = 1000$ it lies in $[0.996, 1.002]$ (Appendix Table 4). Estimated headroom is $\hat{\kappa} \leq 0.024$ (gene_expression), 0.011 (census_income), 0.008 (alphafold) and ≤ 0.002 elsewhere, matching the near-1 ratios. Fig. 1 (left) shows the ratio against n and N: in nearly every real dataset the curve sits at 1.00–1.02 (slightly above because estimating $\hat{g}$ costs a little). Increasing N at $n = 200$ moves the real-data ratios only for gene_expression (0.967 at $N = 5000$, 0.928 at $N = 20000$); all others stay within $[0.99, 1.011]$.

The gene_expression gain has a transparent source: the predictor's slope is $b = 1.15 > 1$, so $\lambda^\star > 1$ and ppi_py clips λ to 1 while AIPW-lin does not. An ablation with unclipped λ (Appendix Table 6) gives 0.975 of the clipped width at $n = 200$ (coverage 94.0 vs 94.2) and exactly 1.000 at $n = 1000$, where the finite-N shrinkage $1/(1 + n/N)$ brings $\lambda^\star$ below 1 again. Unclipping alone thus recovers the gain; no calibration learner is needed. For census_income ($b = 1.38$) the same ablation gives 0.980 at $n = 200$, with coverage already low (91.6% clipped, 90.7% unclipped).

5.2 Synthetic designs: where headroom exists, and how it is realised

Table 1 (bottom) and Fig. 1 (right). S1 (monotone but non-affine calibration, $Y = e^{X_1}$+noise, $F = X_1$), S2 and S3 (the label depends on a covariate the predictor ignores, linearly/non-linearly) and S4 (a pure-noise predictor, informative covariate) have $\hat{\kappa} \approx 0.8$–0.88. At $n = 200$, $N = 5000$ AIPW-auto is 0.550, 0.509, 0.605 and 0.486 of the PPI++ width, with coverage 90.4, 94.7, 94.9 and 96.4%. AIPW-lin helps only on S1 (0.953, thanks to the unclipped slope $b = 1.66$) and not on S2–S4 (1.011), as the theory predicts: the gain requires a calibration class that contains μ, here isotonic

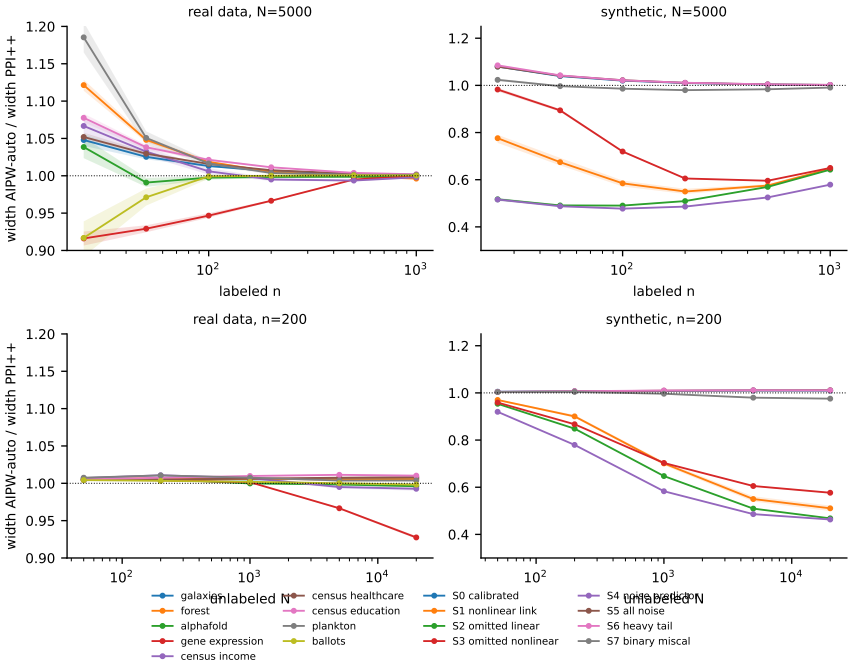


Figure 1: Width of AIPW-auto relative to PPI++ (mean over 1000 replications; bands: paired bootstrap 95% intervals). Real data are essentially flat at 1; synthetic designs with non-affine calibration or informative covariates (S1–S4) fall to 0.49–0.65. The ratio *rises* with n at fixed N because the gap carries the factor $N/(n+N)$ (Eq. 2).

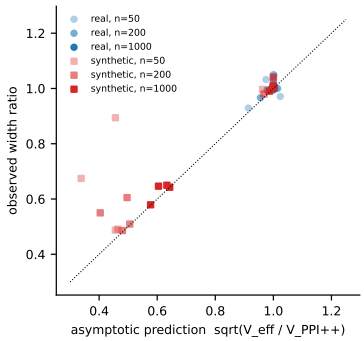


Figure 2: Observed width ratio of AIPW-auto vs. PPI++ against the asymptotic prediction $\sqrt{V_{\text{eff}}/V_{\text{PPI++}}}$ computed from population quantities (Section 4), $N = 5000$, all datasets and designs, $n \in \{50, 200, 1000\}$. Points converge to the diagonal as n grows; at small n synthetic points lie above it (the cost of fitting $\hat{g}$ is not in the formula).

(S1) or covariate-aware (S2–S4); the automatic selection finds it (at $n = 200$ it picks isotonic in 100% of replications for S1, covariate-linear for S2, boosting for S3 and covariate-linear for S4 in 99.8%). Where $\hat{\kappa} = 0$ (S0 calibrated, S5 all noise, S6 heavy tails) the estimators cost 1.0–1.1% extra width, and S7 (binary label, miscalibrated probability, $\hat{\kappa} = 0.013$) gains 2%. S4 also shows PPI++'s protection: its ratio to the classical interval is 1.00 while $\lambda = 1$ PPI is 1.36 wider than PPI++; with an informative covariate PPI++ nonetheless leaves half the width on the table.

Theory versus experiment. Fig. 2 plots the observed against the predicted ratio. At $n = 1000$ the synthetic points lie on the diagonal within 0.05 (S2: predicted 0.643, observed 0.643; S1: 0.604 vs. 0.647). At $n = 200$ the realised gain for S1 and S3 is smaller than predicted (0.55 vs. 0.40; 0.61 vs. 0.50): flexible $\hat{g}$ needs data. The ratio also rises with n at fixed N (Fig. 1, second panel; S2: 0.51 at $n = 200$, 0.57 at $n = 500$, 0.64 at $n = 1000$), exactly the $N/(n+N)$ factor in (2): labels eventually dominate the unlabeled sample and the gap closes.

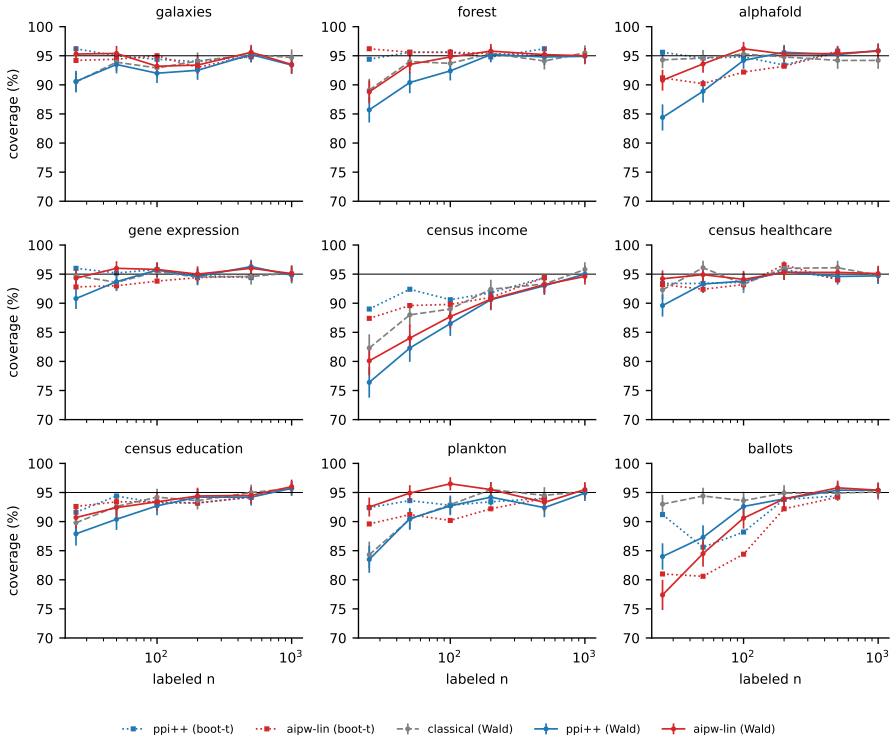


Figure 3: Empirical coverage versus labeled sample size ($N = 5000$). Solid/dashed: Wald; dotted: studentized bootstrap ($B = 399$, 500 replications, error bars for Wald only ± 1.96 s.e.). Nominal 95% (black).

5.3 Validity at small n: all Wald intervals under-cover on real data

Fig. 3 shows that Wald coverage is below nominal for *every* method at small n and does not reach 95% in the heavy-tailed census_income until $n \approx 1000$. Across the nine datasets the mean coverage of PPI++ is 85.9% at $n = 25$, 90.0% at $n = 50$, 92.5% at $n = 100$, 94.0% at $n = 200$ (worst single dataset: 76.4%, 82.3%, 86.5%, 90.6%); AIPW-lin is somewhat better (89.3, 92.1, 93.6, 94.4%) and AIPW-auto close to PPI++ (86.6, 91.2, 92.9, 94.1%). In ballots (predictor error rate 0.9%) a sample of 25–100 labels often contains no or few prediction errors and the variance estimate is too small; AIPW with a fitted $\hat{g}$ is worse there (77.4% at $n = 25$ vs. 84.0%). A comparison of width without coverage would hence flatter whichever method under-covers: at $n = 25$ AIPW-auto is 0.917 of PPI++ on ballots but covers only 77.4%.

Studentized bootstrap intervals (Table 2) raise mean coverage of PPI++ to 93.3% at $n = 25$ and 93.9% at $n = 200$ (Wald: 87.1, 93.5; the 500-replication s.e. per dataset is ≈ 1 point), while the bootstrap AIPW-lin interval is at 90.9% / 93.5% and is in ballots and plankton considerably worse (81% and 90% at $n = 25$). At $n = 200$ the bootstrap AIPW-lin interval is median 1.9% narrower than bootstrap PPI++ (per-dataset range 0.95–0.99), with 0.4 points lower mean coverage; at $n = 25$ it is 13% wider. We therefore see no robust, coverage-adjusted shortening at small n on real data. The classical bootstrap interval is the best-covered (96.0% at $n = 25$) but 1.1–1.4 times wider than PPI++ (median over datasets).

5.4 Failure under labeled/unlabeled shift

A1 (exchangeability) can fail: with labels from one year and unlabeled data from the next, Table 3 shows that the labeled-only interval fails (bias 0.062), PPI with $\lambda = 1$ stays valid (93–95%; the rectifier $Y - F$ transfers across years even though Y does not), while PPI++ and all cross-fitted estimators collapse to 0.0% coverage at $n = 5000$. Any method that uses the labeled data to learn calibration (power tuning, $\hat{g}$) inherits the labeled distribution's bias with a precision that grows with

Table 2: Studentized-bootstrap ($B = 399$, 500 replications, $N = 5000$) coverage (%) and width relative to bootstrap PPI++ (ratio of mean widths), $n = 50$ (top) and $n = 200$ (bottom). Coverage s.e. ≈ 1 point.

dataset	clas.	PPI++	AIPW-lin	clas. width	AIPW-lin width
galaxies	96.0	95.0	94.4	1.382	0.963
forest	98.0	95.6	95.6	0.935	1.126
alphafold	97.6	94.6	90.2	0.843	0.947
gene_expression	95.0	95.2	93.0	2.972	0.848
census_income	92.8	92.4	89.6	1.093	1.019
census_healthcare	95.6	93.4	92.4	1.358	0.989
census_education	94.6	94.4	93.4	1.209	0.999
plankton	97.8	93.6	91.2	0.980	1.614
ballots	96.6	85.6	80.6	3.311	0.891
galaxies	94.6	94.0	93.0	1.423	0.981
forest	94.2	94.8	95.4	1.150	0.991
alphafold	95.0	93.4	93.2	1.584	0.956
gene_expression	94.4	94.8	94.4	2.887	0.950
census_income	93.2	91.8	91.0	1.168	0.987
census_healthcare	95.2	95.8	96.6	1.369	0.993
census_education	93.2	93.2	93.2	1.247	0.987
plankton	94.8	93.4	92.2	1.672	0.954
ballots	97.0	93.8	92.2	3.475	0.971

Table 3: Plankton: labeled data from 2013, unlabeled from 2014, target the 2014 mean label (0.071; 2013 mean 0.134). Coverage (%), $N = 5000$, 1000 replications.

n	classical	PPI	PPI++	AIPW-lin	AIPW-auto
100	53.0	95.4	95.2	96.7	96.7
500	1.0	94.7	94.7	72.1	72.1
5000	0.0	93.0	0.0	0.0	0.0

n. This is a conditional benefit: efficiency gains from fitted calibration are traded against robustness to shift, and both PPI++ and our estimators are on the fragile side.

6 Discussion and Limitations

Answer to the question. For a population mean, with a predictor that is calibrated up to an affine map and no covariate information the predictor missed, the optimally tuned PPI++ interval is already (asymptotically) as short as any regular estimator permits, and on nine real datasets we measure at most a 3.3% ($n = 200$, gene_expression) reduction, explained by an un-clipped λ. The interval can be substantially shortened (here by up to $\approx 50\%$) only when $\mathbb{E}[Y \mid Z]$ departs from an affine function of the prediction, e.g. non-linear calibration or unused covariates; the gain scales with $N/(n+N)$, needs enough labels for $\hat{g}$ (our selected learners pay off from $n \approx 100$–200), and disappears with a calibrated predictor. Practitioners can estimate $\hat{\kappa}$ from the labeled data (cross-validated residual variances) before choosing a method.

Limitations. (i) *Target.* Only the mean. Regression coefficients, quantiles and other M-estimands have analogous efficient influence functions (conditional expectation of the score given Z) but we did not test them; for them headroom may differ. (ii) *Learner class.* $\hat{\kappa}$ comes from four learner families under cross-validation; richer learners might find more headroom in the real datasets, so "at most 2.4%" is a statement about these learners. (iii) *Finite samples.* Proposition 3 is asymptotic and its sketch not a full proof; at $n \leq 100$ real-data coverage falls below nominal for all methods and our bootstrap fix is partial. We did not try other corrections (Edgeworth, exact binomial); t-quantile versions are stored in the result files but not analysed. (iv) *Sampling design.* Datasets are resampled with replacement from a finite file, so results are about the empirical distribution of each file; small files (forest, ballots) are heavily bootstrapped at $N = 5000$. The ballots population is the 1027 labeled ballots only. (v) *Exchangeability* is assumed; Section 5.4 is a single shift example. (vi) The classical

and PPI baselines use the `ppi-python` standard errors ($\mathrm{ddof} = 0$), our estimators $\mathrm{ddof} = 1$, a 2% width difference at $n = 25$ that we did not remove. (vii) We did not study quantile/binary-optimal intervals such as Wilson or exact methods, which may be shorter at small n for rare-error binary labels. (viii) Compute was small (CPU only) and reproducibility is in the supplementary code.

Broader impact. The work is methodological and uses public data; the main practical risk is overconfidence from intervals reported in the small-label regime where we find under-coverage.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023a. arXiv:2301.09633.

Anastasios N. Angelopoulos, John C. Duchi, and Tijana Zrnic. PPI++: Efficient prediction-powered inference. *arXiv preprint arXiv:2311.01453*, 2023b.

Peter J. Bickel, Chris A. J. Klaassen, Ya'acov Ritov, and Jon A. Wellner. *Efficient and Adaptive Estimation for Semiparametric Models*. Johns Hopkins University Press, 1993.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. arXiv:1608.00060.

Adam Fisch, Joshua Maynez, R. Alex Hofer, Bhuwan Dhingra, Amir Globerson, and William W. Cohen. Stratified prediction-powered inference for hybrid language model evaluation. *arXiv preprint arXiv:2406.04291*, 2024.

Jiacheng Miao, Xinran Miao, Yixuan Wu, Jiwei Zhao, and Qiongshi Lu. Assumption-lean and data-adaptive post-prediction inference. *arXiv preprint arXiv:2311.14220*, 2023.

Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427): 846–866, 1994.

Anastasios A. Tsiatis. *Semiparametric Theory and Missing Data*. Springer, 2006.

Anru Zhang, Lawrence D. Brown, and T. Tony Cai. Semi-supervised inference: General theory and estimation of means. *The Annals of Statistics*, 47(5):2538–2566, 2019.

Tijana Zrnic and Emmanuel J. Candès. Cross-prediction-powered inference. *arXiv preprint arXiv:2309.16598*, 2023.

Tijana Zrnic and Emmanuel J. Candès. Active statistical inference. *arXiv preprint arXiv:2403.03208*, 2024.

A Proofs

Proposition 1. Write $\hat{\theta}_g = \sum_{i \le n} \left[\frac{Y_i}{n} + (\frac{1}{n+N} - \frac{1}{n}) g(Z_i)\right] + \frac{1}{n+N} \sum_{j \le N} g(Z'_j)$; the two sums are independent sums of i.i.d. terms, $c = \frac{1}{n+N}$, $a = \frac{1}{n}$. Then $\mathrm{Var} = n[a^2 \mathrm{Var} Y + 2a(c-a)\mathrm{Cov}(Y, g) + (c-a)^2 \mathrm{Var} g] + N c^2 \mathrm{Var} g$. With $c - a = -\frac{N}{n(n+N)}$: $n(c-a)^2 + Nc^2 = \frac{N}{n(n+N)}$ and $2na(c-a) = -\frac{2N}{n(n+N)}$, so $\mathrm{Var} = \frac{\mathrm{Var} Y}{n} + \frac{N}{n(n+N)}(\mathrm{Var} g - 2\mathrm{Cov}(Y, g)) = \frac{\mathrm{Var} Y}{n+N} + \frac{N}{n(n+N)} \mathrm{Var}(Y - g)$, using $\frac{1}{n} - \frac{N}{n(n+N)} = \frac{1}{n+N}$. Unbiasedness: $\mathbb{E} g$ cancels between the terms. The minimum over g of $\mathrm{Var}(Y - g(Z))$ is $\mathbb{E}\mathrm{Var}(Y \mid Z)$ at $g = \mu$.

Proposition 2. $\mathrm{Var}(Y - \lambda F)/n + \lambda^2 \mathrm{Var} F / N$ is a quadratic in λ minimized at $\lambda^\star$ with value $\frac{1}{n}[\mathrm{Var} Y - \mathrm{Cov}(Y, F)^2 / (\mathrm{Var} F(1 + n/N))] = \frac{1}{n}[\mathrm{Var} Y - \frac{N}{n+N} b^{\star 2} \mathrm{Var} F]$. By Prop. 1, $\min_b \mathrm{Var}(\hat{\theta}_{bF}) = \frac{\mathrm{Var} Y}{n+N} + \frac{N}{n(n+N)}(\mathrm{Var} Y - b^{\star 2} \mathrm{Var} F)$, which equals the same expression. For

(2): $\mathrm{Var}(Y - b^\star F) = \mathbb{E}\mathrm{Var}(Y \mid Z) + \mathrm{Var}(\mu - b^\star F)$ because $Y - \mu \perp (\mu - b^\star F)$; subtract V_{eff}.

Corollary 1. As $N/n \to \infty$, $nV_{\mathrm{PPI++}} \to \mathrm{Var}(Y - b^\star F)$ and $nV_{\mathrm{eff}} \to \mathbb{E}\mathrm{Var}(Y \mid Z)$, whose ratio is $1 - \kappa$. (i) μ affine in F implies $\mu = a + b^\star F$ and the gap vanishes. (iii) $\lambda \mapsto \mathrm{Var}(\hat{\theta}^{\mathrm{PPI}}_\lambda)$ is strictly convex, so clipping an interior minimizer is strictly suboptimal.

Proposition 3 (sketch). Let $\hat{\theta}^*$ denote (1) with $g = g^*$. For fold k, conditional on the training folds, $\hat{\theta} - \hat{\theta}^*$ is a sum over folds of $(\mathrm{mean}_{\mathrm{all}} - \mathrm{mean}_{\mathrm{lab}})(\hat{g}^{(-k)} - g^*)$, a centred difference of sample means of one fixed function with variance $\leq C \, \|\hat{g}^{(-k)} - g^*\|^2_{L_2} \, (1/n) = o_p(1/n)$ (the usual double machine learning argument [Chernozhukov et al., 2018]); the unlabeled average over the K fold models adds only terms of the same type. Then $\hat{\theta}^*$ is asymptotically normal by the Lindeberg CLT applied to the two independent samples with variance from Prop. 1, and $\hat{V}$ is consistent because the empirical variances of the plug-in terms converge to those of the g^*-terms by the same L_2 argument. A full proof with explicit moment conditions is not given.

B Synthetic designs

$X_1, X_2, F_0 \sim \mathcal{N}(0,1)$, $\varepsilon \sim \mathcal{N}(0,1)$ independent; W are the covariates available to the analyst. **S0:** $Y = X_1 + \varepsilon$, $F = X_1$. **S1:** $Y = e^{X_1} + 0.5\varepsilon$, $F = X_1$ ($\theta = e^{1/2}$). **S2:** $Y = X_1 + X_2 + 0.5\varepsilon$, $F = X_1$, $W = X_2$. **S3:** $Y = X_1 + 1.5\sin(2X_2) + 0.5\varepsilon$, $F = X_1$, $W = X_2$. **S4:** $Y = X_1 + 0.5\varepsilon$, $F = F_0$ (noise), $W = X_1$. **S5:** $Y = X_1 + \varepsilon$, $F = F_0$, W two independent noise columns. **S6:** $Y = X_1 + t_3$, $F = X_1$. **S7:** $Y \sim \mathrm{Bernoulli}(\sigma(2X_1))$, $F = \sigma(X_1)$ ($\theta = 0.5$). Population constants ($\hat{\kappa}$, b) are from 2×10^5 draws.

C Additional results

Table 4: As Table 1 for $n = 1000$ (top: real; bottom: synthetic).

				coverage (%)					width / PPI++			
dataset	b	$\hat{\kappa}$	bound	clas.	PPI	PPI++	lin	auto	clas.	PPI	lin	auto
galaxies	0.98	0.000	1.000	94.7	93.9	93.4	93.5	93.4	1.358	1.021	1.001	1.001
forest	0.90	0.000	1.000	95.5	94.6	94.9	95.0	94.4	1.148	1.018	1.002	0.996
alphafold	1.03	0.008	0.996	94.2	96.0	95.9	95.8	95.2	1.498	1.015	1.001	0.999
gene_expression	1.15	0.024	0.988	95.2	95.3	94.8	95.1	95.1	2.099	1.002	1.000	1.000
census_income	1.38	0.011	0.994	95.8	95.0	95.0	94.6	94.5	1.199	1.000	1.000	0.998
census_healthcare	0.98	0.001	0.999	94.8	94.6	94.7	95.1	95.2	1.308	1.017	1.002	1.001
census_education	0.97	0.001	1.000	95.7	96.3	95.7	96.0	96.2	1.220	1.013	1.002	1.002
plankton	0.84	0.000	1.000	95.2	95.3	94.9	95.5	95.5	1.548	1.116	1.001	1.001
ballots	0.98	0.000	1.000	95.1	94.9	95.4	95.4	95.4	2.253	1.101	1.000	1.000
S0_calibrated	1.00	0.000	1.000	95.0	95.4	95.9	95.8	95.8	1.309	1.014	1.002	1.002
S1_nonlinear_link	1.66	0.884	0.341	94.3	94.4	94.4	94.6	93.6	1.330	1.000	0.974	0.647
S2_omitted_linear	1.00	0.798	0.449	94.6	94.3	94.7	94.7	94.6	1.260	1.012	1.002	0.643
S3_omitted_nonlinear	1.00	0.806	0.441	95.1	95.5	95.0	95.0	94.5	1.241	1.011	1.002	0.650
S4_noise_predictor	-0.00	0.800	0.447	95.1	95.5	95.2	95.0	95.2	1.000	1.402	1.002	0.579
S5_all_noise	-0.01	0.000	1.000	94.3	95.3	94.3	94.3	94.4	1.000	1.264	1.002	1.002
S6_heavy_tail	1.00	0.000	1.000	95.4	95.6	95.4	95.8	95.7	1.129	1.006	1.002	1.002
S7_binary_miscal	1.49	0.013	0.993	94.2	94.5	94.5	93.8	94.0	1.203	1.000	0.993	0.991

D Reproducibility and notes

Code: `code/` (run order and commands in `README.md`); data: `ppi-python` 0.2.3 example sets (downloaded via `ppi_py.datasets.load_dataset`); Python 3.11, numpy 2.4.6, scipy 1.17.1, pandas 3.1.0, scikit-learn 1.9.1, statsmodels 0.15.0 (installed, unused), matplotlib 3.11.2. Main runs took about 42 min (real), 41 min (synthetic), 24 min (bootstrap) wall-clock on a shared 40-core machine (observed per-process CPU share suggests an effective throughput of roughly 4–5 cores). No GPU, no paid API. No part of this paper's analysis depends on the plankton 2014 labels except Section 5.4. Table 1/4/5: `run_experiments.py`, `analysis.py`, `theory.py`, `make_report.py`; Fig. 1, 2: same; Fig. 3, Table 2: `run_boot.py`; Table 3: `run_shift.py`; Table 6: `run_noclip.py`.

Table 5: As Table 1 for $n = 50$.

dataset	b	$\hat{\kappa}$	bound	coverage (%) clas.	PPI	PPI++	lin	auto	width / PPI++ clas.	PPI	lin	auto
galaxies	0.98	0.000	1.000	93.9	95.3	93.5	95.4	94.4	1.486	0.996	1.040	1.025
forest	0.90	0.000	1.000	94.0	91.9	90.4	93.5	92.2	1.192	1.007	1.064	1.048
alphafold	1.03	0.008	0.996	94.6	92.7	88.9	93.6	90.2	1.723	0.982	1.030	0.991
gene_expression	1.15	0.024	0.988	93.6	94.4	93.7	96.0	95.7	3.220	0.976	0.933	0.929
census_income	1.38	0.011	0.994	88.0	83.4	82.3	84.0	82.9	1.270	0.998	1.055	1.032
census_healthcare	0.98	0.001	0.999	96.1	94.0	93.3	94.9	93.8	1.413	1.000	1.044	1.030
census_education	0.97	0.001	1.000	92.6	91.8	90.4	92.4	92.0	1.291	1.006	1.056	1.038
plankton	0.84	0.000	1.000	90.4	96.8	90.5	94.9	94.9	1.893	1.005	1.051	1.051
ballots	0.98	0.000	1.000	94.4	84.3	87.3	84.5	84.5	5.397	0.950	0.971	0.971
S0_calibrated	1.00	0.000	1.000	93.9	94.4	94.2	95.2	95.1	1.409	0.998	1.044	1.039
S1_nonlinear_link	1.66	0.884	0.341	89.0	84.9	84.5	88.4	86.4	1.445	0.997	1.005	0.674
S2_omitted_linear	1.00	0.798	0.449	94.2	93.9	94.0	94.7	95.0	1.340	0.999	1.048	0.491
S3_omitted_nonlinear	1.00	0.806	0.441	93.0	93.9	93.3	94.0	93.2	1.309	1.001	1.046	0.894
S4_noise_predictor	-0.00	0.800	0.447	93.4	93.6	93.3	94.2	94.6	1.005	1.359	1.051	0.488
S5_all_noise	-0.01	0.000	1.000	95.8	95.0	95.1	95.9	95.0	1.005	1.232	1.051	1.041
S6_heavy_tail	1.00	0.000	1.000	95.2	95.1	94.8	95.4	95.0	1.183	1.002	1.048	1.042
S7_binary_miscal	1.49	0.013	0.993	93.1	93.5	93.5	93.9	92.4	1.234	1.000	1.006	0.997

Table 6: PPI++ with λ clipped to $[0, 1]$ (as in `ppi_py`) versus unclipped; $N = 5000$, 2000 replications. Columns: dataset, n, coverage clipped, coverage unclipped (%), width ratio unclipped/clipped.

dataset	n	cov. clipped	cov. unclipped	width ratio
S1_nonlinear_link	200	91.8	91.5	0.924
S1_nonlinear_link	1000	92.9	93.0	0.969
S7_binary_miscal	200	94.5	94.5	0.974
S7_binary_miscal	1000	94.4	94.7	0.991
alphafold	200	93.8	93.4	1.008
alphafold	1000	94.6	94.6	1.000
census_income	200	91.6	90.7	0.980
census_income	1000	94.5	94.1	0.996
gene_expression	200	94.2	94.0	0.975
gene_expression	1000	95.4	95.4	1.000

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